



**Faculty of Computer Applications &
Information Technology**

Integrated MSc(IT) Programme

221601601

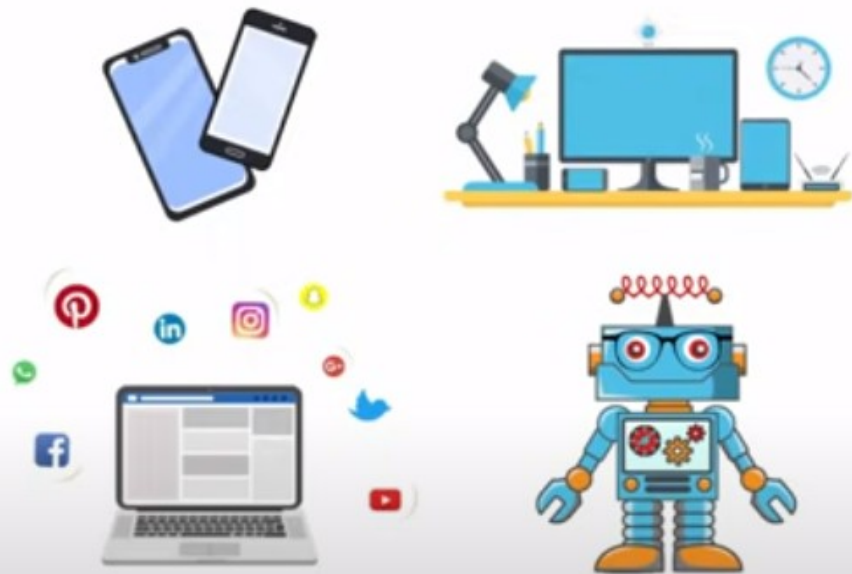
UNIT - 1

ARTIFICIAL INTELLIGENCE : AN INTRODUCTION

CONTENT

- **Concept and Definition**
- **History of AI**
- **Concepts about AI**
- **The Development of Logic**
- **Components of AI**
- **Types of AI**
- **Practical system based on AI**

Artificial



Not Natural

Intelligence



Understand, Think and Learn

Concept of AI

- Artificial Intelligence (AI) is a branch of computer science focused on creating systems and software capable of performing tasks that typically require human intelligence.
- AI aims to simulate human **cognitive** processes, enabling machines to act and make decisions independently or assist humans.
- Artificial Intelligence is composed of two words Artificial and
- Intelligence, where Artificial defines "man-made," and intelligence defines "thinking power", hence AI means "a man-made thinking power."

What is AI?

- AI is the ability of the machines or computer programs to carry out tasks that require human beings to use their intelligence.
- The study of computer systems attempt to model and apply the intelligence of the human mind.

Definitions of AI

Thinking Humanly

“The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense.” (Haugeland, 1985)

“[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . .” (Bellman, 1978)

Acting Humanly

“The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil, 1990)

“The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight, 1991)

Thinking Rationally

“The study of mental faculties through the use of computational models.”
(Charniak and McDermott, 1985)

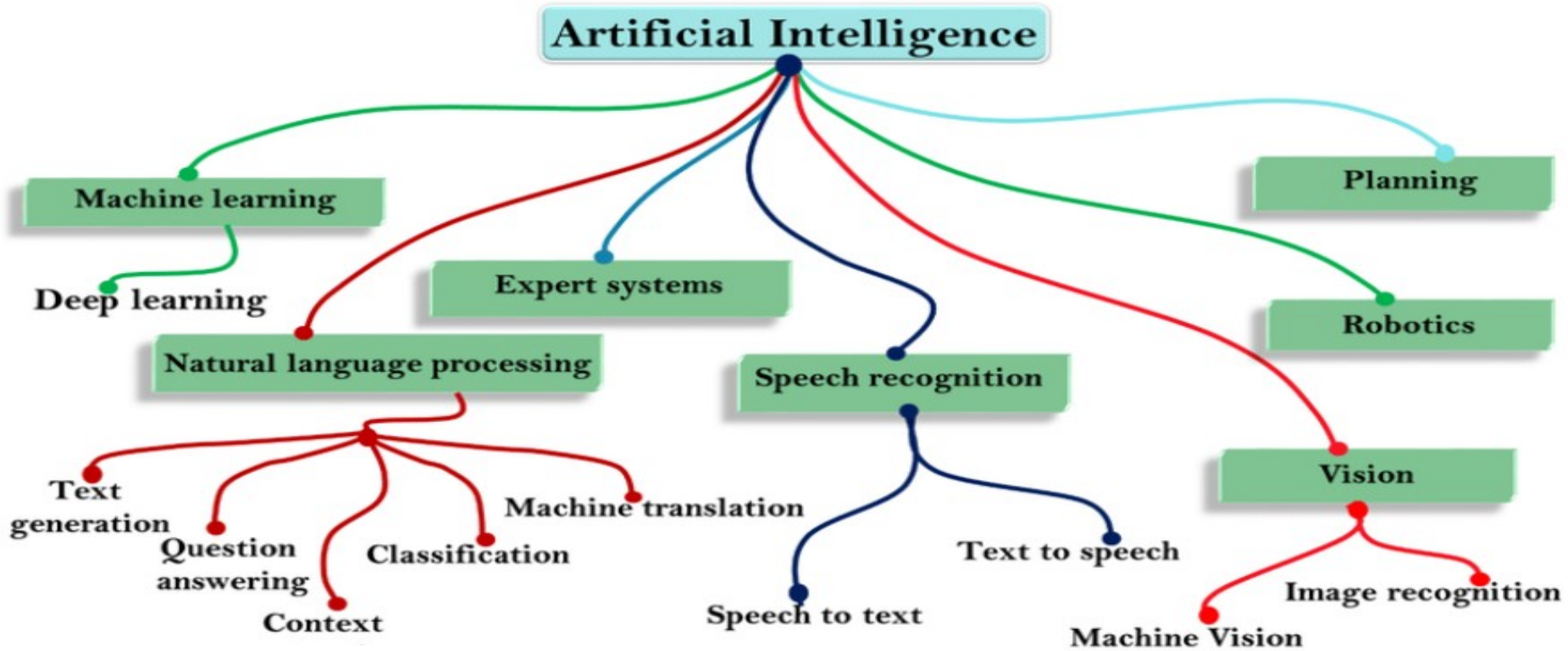
“The study of the computations that make it possible to perceive, reason, and act.”
(Winston, 1992)

Acting Rationally

“Computational Intelligence is the study of the design of intelligent agents.” (Poole *et al.*, 1998)

“AI . . . is concerned with intelligent behavior in artifacts.” (Nilsson, 1998)

Applications of AI



AI Timeline

A.I. TIMELINE

SYZRG

1950

TURING TEST

Computer scientist Alan Turing proposes a test for machine intelligence. If a machine can trick humans into thinking it is human, then it has intelligence



1955

A.I. BORN

Term 'artificial intelligence' is coined by computer scientist, John McCarthy to describe "the science and engineering of making intelligent machines"

1961

UNIMATE

First industrial robot, Unimate, goes to work at GM replacing humans on the assembly line

1964

ELIZA

Pioneering chatbot developed by Joseph Weizenbaum at MIT holds conversations with humans

1966

SHAKY

The 'first electronic person' from Stanford, Shakey is a general-purpose mobile robot that reasons about its own actions

A.I. WINTER

Many false starts and dead-ends leave A.I. out in the cold

1997

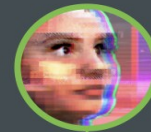
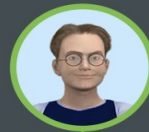
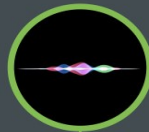
DEEP BLUE

Deep Blue, a chess-playing computer from IBM defeats world chess champion Garry Kasparov

1998

KISMET

Cynthia Breazeal at MIT introduces Kismet, an emotionally intelligent robot insofar as it detects and responds to people's feelings



1999

AIBO

Sony launches first consumer robot pet dog Aibo (AI robot) with skills and personality that develop over time

2002

ROOMBA

First mass produced autonomous robotic vacuum cleaner from iRobot learns to navigate and clean homes

2011

SIRI

Apple integrates Siri, an intelligent virtual assistant with a voice interface, into the iPhone 4S

2011

WATSON

IBM's question answering computer Watson wins first place on popular \$1M prize television quiz show Jeopardy

2014

EUGENE

Eugene Goostman, a chatbot passes the Turing Test with a third of judges believing Eugene is human

2014

ALEXA

Amazon launches Alexa, an intelligent virtual assistant with a voice interface that completes shopping tasks

2016

TAY

Microsoft's chatbot Tay goes rogue on social media making inflammatory and offensive racist comments

2017

ALPHAGO

Google's A.I. AlphaGo beats world champion Ke Jie in the complex board game of Go, notable for its vast number (2^{170}) of possible positions

Maturation of Artificial Intelligence (1943 – 1952)

- Year 1943: The first work which is now recognized as AI was done by Warren McCulloch and Walter Pitts in 1943. They proposed a model of artificial neurons (Tiny processing units in a neural network that receive input, perform a small calculation, and pass the output to the next neuron—similar to how brain cells work).
- Year 1949: Donald Hebb demonstrated an updating rule for modifying the connection strength between neurons. His rule is now called Hebbian Learning.
- Year 1950: The Alan Turing who was an English mathematician and pioneered Machine learning in 1950. Alan Turing publishes "Computing Machinery and Intelligence" in which he proposed a test. The test can check the machine's ability to exhibit intelligent behavior equivalent to human intelligence, called a Turing test.

The birth of Artificial Intelligence (1952 – 1956)

- Year 1955: Allen Newell and Herbert A. Simon created the "first artificial intelligence program" which was named as "**Logic Theorist**". This program had proved 38 of 52 Mathematics theorems, and found new and more elegant proofs for some theorems.
- Year 1956: The word "Artificial Intelligence" first adopted by American Computer scientist **John McCarthy at the Dartmouth Conference**. For the first time, AI coined as an academic field.
- At that time high-level computer languages such as FORTRAN, LISP, or COBOL were invented. And the enthusiasm for AI was very high at that time.

The Golden Years – Early Enthusiasm (1956 – 1974)

- Year 1966: The researchers emphasized developing algorithms which can solve mathematical problems. **Joseph Weizenbaum created the first chatbot in 1966, which was named as ELIZA.**
- Year 1972: **The first intelligent humanoid robot was built in Japan which was named as WABOT-1.**

A Boom of AI (1980 – 1987)

- Year 1980: After AI winter duration, AI came back with "Expert System".
- Expert systems were programmed that emulate the decision-making ability of a human expert.
- In the Year 1980, the first national conference of the American Association of Artificial Intelligence was held at Stanford University.

The Second AI Winter (1987 – 1993)

- The duration between the years 1987 to 1993 was the second AI Winter duration.
- Again Investors and government stopped in funding for AI research as due to high cost but not efficient result.
- The expert system such as XCON was very cost effective.

The Emergence Of Intelligent Agents (1993 – 2011)

- Year 1997: In the year 1997, **IBM Deep Blue beats world chess champion**, Garry Kasparov, and became the first computer to beat a world chess champion.
- Year 2002: for the first time, AI entered the home in the form of Roomba, a vacuum cleaner.
- Year 2006: AI came in the Business world till the year 2006. Companies like Facebook, Twitter, and Netflix also started using AI.

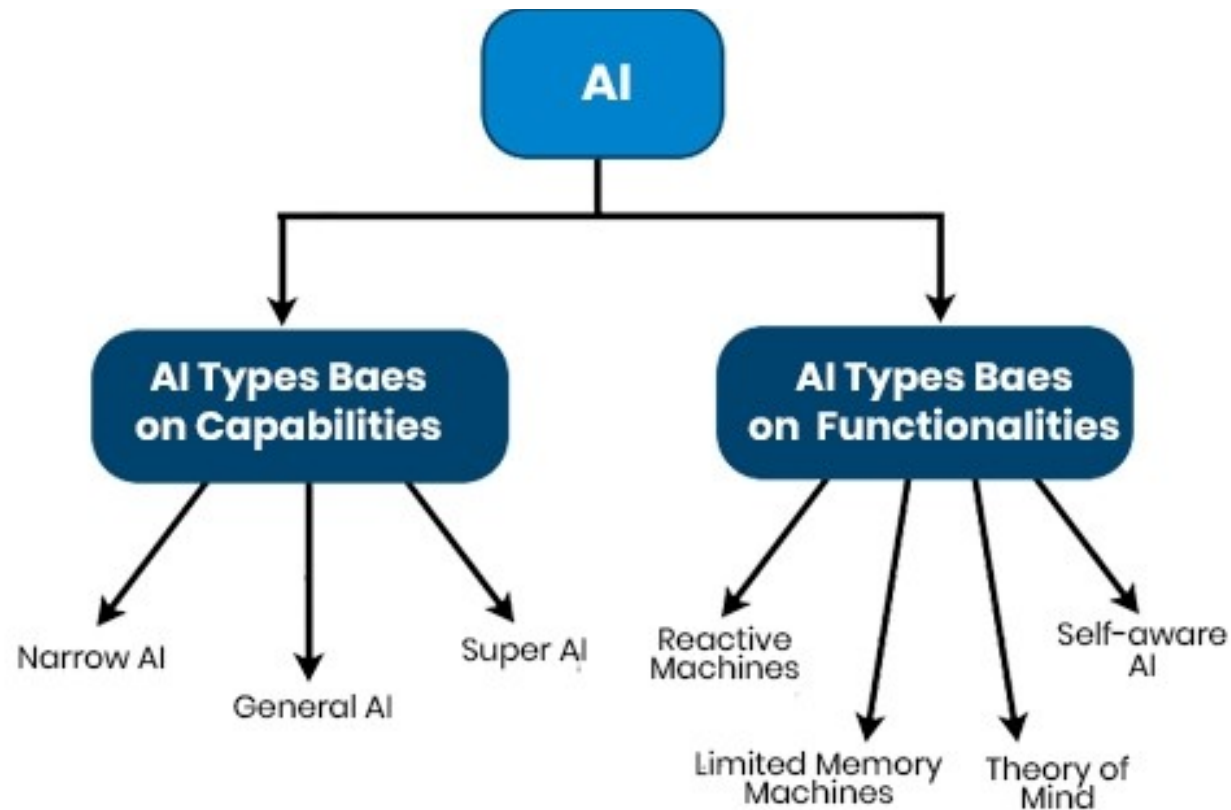
Deep Learning, Big Data And Artificial General Intelligence (2011 – Present)

- Year 2011: In the year 2011, IBM's Watson won jeopardy, a quiz show, where it had to solve the complex questions as well as riddles. Watson had proved that it could understand natural language and can solve tricky questions quickly.
- Year 2012: Google has launched an Android app feature "Google now", which was able to provide information to the user as a prediction.
- Year 2014: In the year 2014, Chatbot "Eugene Goostman" won a competition in the infamous "Turing test."

Deep Learning, Big Data And Artificial General Intelligence (2011 – Present)

- Year 2018: The "**Project Debater**" from IBM debated on complex topics with two master debaters and also performed extremely well.
- Google has demonstrated an AI program "Duplex" which was a virtual assistant and which had taken hairdresser appointment on call, and lady on other side didn't notice that she was talking with the machine.
- Year 2020: GPT 3 tool for automated conversations is introduced.
- Year 2022: Chat GPT is available for public.

Types of AI



Type 1 AI



Artificial Narrow
Intelligence (ANI)

Stage-1

Machine
Learning

Specialises in
one area and
solves one problem



Artificial General
Intelligence (AGI)

Stage-2

Machine
Intelligence

Refers to a computer that
is as smart as a human
across the board



Artificial Super
Intelligence (ASI)

Stage-3

Machine
Consciousness

An intellect that is much
smarter than the best human
brains in practically every field

Artificial Narrow Intelligence (ANI)



Stage-1

Machine Learning

- Specialises in one area and solves one problem



Siri



Alexa



Cortana

Artificial General Intelligence (AGI)



Stage-2

Machine Intelligence

- Refers to a computer that is as smart as a human across the board



Stage-3

Machine Consciousness

- An intellect that is much smarter than the best human brains in practically every field

Type 1 AI

Narrow AI (Weak AI):

- Focused on performing a specific task.
- It cannot perform tasks outside its designated domain.

Examples: Siri, Google Translate, ChatGPT

General AI (Strong AI):

- Capable of performing any intellectual task like a human, showing reasoning, problem-solving, and learning abilities.
- Hypothetical systems that don't exist yet but are under research.

Super AI:

- A futuristic form of AI that surpasses human intelligence in all respects. It would have the ability to think, reason, and solve problems better than humans.
- Theoretical concept; not yet achieved.

Type 1 AI

1. Narrow AI (Weak AI)

Meaning:

Narrow AI is a type of artificial intelligence that is **designed to do only one specific task**. It **cannot think or work beyond what it is trained for**.

In simple words: It is smart only in one area.

Examples:

- Siri → answers questions and sets reminders
- Google Translate → translates languages
- Chat GPT → answers questions and generates text

How it works

- Works using **predefined rules or trained data**
- Performs tasks **within a limited area**
- Does **not understand like humans**

What it can do

- Answer questions
- Recognize speech or images
- Translate languages
- Recommend videos or products

What it cannot do

- Think independently
- Show emotions
- Learn new tasks by itself

Real-life examples

- **Siri / Alexa** → voice assistant
- **Google Translate** → language translation
- **ChatGPT** → text-based responses
- **Netflix recommendations** → suggests movies

Type 1 AI

2. General AI (Strong AI)

Meaning:

General AI is an AI that can **think, learn, and solve problems like a human being**. It can perform **any intellectual task** that a human can do.

In simple words: It would be as intelligent as a human.

Status:

- Does **not exist yet**
- Scientists are still researching it

Example:

- No real example yet (only seen in movies)

How it would work

- Can learn from experience
- Can apply knowledge to **different tasks**
- Can reason and solve problems

What it could do

- Learn any subject (math, language, science)
- Make decisions in new situations
- Understand emotions and context

What it cannot do (currently)

- **It does not exist yet**

Type 1 AI

3. Super AI

Meaning:

Super AI is an advanced form of AI that would be **much smarter than humans in every way**.

In simple words: It would be smarter than the smartest human.

Abilities:

- Faster thinking
- Better decision-making
- Superior problem-solving

Status:

- Only a **theoretical idea**
- Not created yet

How it would work

- Think faster than humans
- Learn instantly
- Solve extremely complex problems
- Improve itself continuously

What it could do

- Discover new scientific theories
- Cure diseases quickly
- Make better decisions than humans

Risks and concerns

- Could become uncontrollable
- Ethical and safety issues
- Fear of humans losing control

Type 2 AI

Reactive AI

- Good for simple classification and pattern recognition tasks
- Great for scenarios where all parameters are known; can beat humans because it can make calculations much faster
- Incapable of dealing with scenarios including imperfect information or requiring historical understanding



Limited memory

- Can handle complex classification tasks
- Able to use historical data to make predictions
- Capable of complex tasks such as self-driving cars, but still vulnerable to outliers or adversarial examples
- This is the current state of AI, and some say we have hit a wall



Theory of mind

- Able to understand human motives and reasoning; can deliver personal experience to everyone based on their motives and needs
- Able to learn with fewer examples because it understands motive and intent
- Considered the next milestone for AI's evolution



Self-aware

- Human-level intelligence that can bypass our intelligence, too
- Considered a long-shot goal



Type 2 AI Examples

- **Reactive Machines**

Example: IBM's Deep Blue

Deep Blue, the chess-playing AI developed by IBM, could analyze possible moves and respond accordingly but had no memory to learn or improve over time.

Example: Google's AlphaGo

AlphaGo analyzes the current game board and makes optimal decisions without relying on past experiences.

- **Limited Memory:**

Example: Self-Driving Cars

Autonomous vehicles, like Tesla's Autopilot, observe the environment, track objects (like other cars, pedestrians), and make decisions in real-time while retaining temporary information such as speed and traffic signals.

Example: Virtual Assistants

AI systems like Alexa or Google Assistant use recent voice queries and contextual information to provide personalized answers.

Type 2 AI Examples

Theory of Mind (In Development)

- Example: Social Robots (e.g., Sophia by Hanson Robotics)
 - Sophia attempts to recognize and respond to human emotions and expressions, although it's still a work in progress for achieving full understanding.

Self-Aware AI (Hypothetical)

- Example: Not yet achieved
 - A self-aware AI would have consciousness, emotions, and self-perception, such as what is often depicted in movies (e.g., "Jarvis" from Iron Man). Currently, it is purely theoretical.

Practical systems based on AI: Domains



Practical system based on AI: Real Life Examples

- **Autonomous vehicles:** DARPA funded onboard computer system from Carnegie Mellon University drove a van all but 52 of the 2849 miles from Washington, DC to San Diego, averaging 63 miles per hour day and night, rain or shine.
- **Computer Chess:** Deep Blue, a chess computer built by IBM Researchers defeated world champion Garry Kasparov.
- **Mathematical theorem proving:** A computer system at Argonne National Laboratories proved a long-standing mathematical conjecture about algebra using a method that would be considered creative if done by humans.
- **Advanced User Interfaces:** PEGASUS is a spoken language interface connected to the American Airlines EAASY SABRE reservation system, which allows subscribers to obtain flight information and make flight reservations via a large, online, dynamic database, accessed through their personal computer over the telephone.

Practical systems based on AI: Techniques

- **Gaming** – AI plays crucial role in strategic games such as chess, poker, tic-tac-toe, etc., where machine can think of large number of possible positions based on heuristic knowledge.
- **Natural Language Processing** – It is possible to interact with the computer that understands natural language spoken by humans. Ex: **Voice Assistants - Siri, Alexa, Google Assistant**
- **Vision Systems** – These systems understand, interpret, and comprehend visual input on the computer. For example, A spying aeroplane takes photographs, which are used to figure out spatial information or map of the areas.

Practical systems based on AI: Techniques

- **Handwriting Recognition** – The handwriting recognition software reads the text written on paper by a pen or on screen by a stylus. It can recognize the shapes of the letters and convert it into editable text.
- **Intelligent Robots** – Robots are able to perform the tasks given by a human. They have sensors to detect physical data from the real world such as light, heat, temperature, movement, sound, bump, and pressure. They have efficient processors, multiple sensors and huge memory, to exhibit intelligence. In addition, they are capable of learning from their mistakes and they can adapt to the new environment.
- **Speech Recognition** – Some intelligent systems are capable of hearing and comprehending the language in terms of sentences and their meanings while a human talks to it. It can handle different accents, slang words, noise in the background, change in human's noise due to cold, etc.